

QPS Line S Recovery — Technical Analysis Report

RTM-292 Pressure Analysis & LOOP Boil-off Transient

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1 Scope

This note answers the RTM-292 question: *for a given helium inflow rate into Line S during a LOOP (Loss of Offsite Power) event, how long before the Line S pressure exceeds its 1.30 bar ceiling, and what recovery suction rate is required?*

A secondary question is addressed: how long does the C30 liquid helium inventory last (boil-off transient), and what peak flow must the recovery system handle?

1.1 Requirements traceability

Requirement	Statement	Binding value
RTM-261	Accept ≥ 200 g/s normal / ≥ 100 g/s abnormal return, no loss	100 g/s HP suction required
RTM-292	Line S pressure within limits during inflow	$P \leq 1.30$ bar
OFFER-22	Applicant recovery strategy, max accepted flow	112 g/s single HP max

1.2 Key resolved parameters

Parameter	Value	Source	Confidence
Line S binding ceiling P_{limit}	1.30 bar	He-line-safety (p_fullop)	High
Nominal operating pressure P_{nom}	1.10 bar	RTM-292	High
Allowable ΔP	0.20 bar	$P_{\text{limit}} - P_{\text{nom}}$	High
Vessel relief set	1.50 bar	He_line_safety	— (excluded from Line S analysis)
Effective gas volume V_{eff}	3.12 m³	DN150 geometry	Medium
HP compressor max suction	112 g/s	Confirmed (350 kW diesel)	High
C30 LHe inventory	391.5 kg	2700 L $\times$ 0.145 kg/L	Medium

Note on 1.50 bar: the vessel relief set applies to the closed isochoric liquid vessel only. It is **not** a Line S structural limit and is excluded from this analysis.

1.3 V_eff breakdown

Segment	Volume (m ³)	Temperature
C30 bath vapour space (10% of 2900 L)	0.289	4 K
Cold Line S — 90 m DN150 (A = 0.01767 m ²)	1.590	4 K (cold)
Warm Line S — 60 m DN150 (A = 0.01767 m ²)	1.060	300 K
QVE user volume	0.185	4 K
Total V_eff	3.124 m³	—

Legacy value of 120 m³ (inherited from an unrelated storage-vessel prompt) is **retired** — it was 38× too large and made the calculated t_available meaningless.

2 Governing equations

Equation 1 — Ideal gas pressure rise rate

The Line S effective volume V_eff receives a net helium mass flow m_net. Under the isothermal ideal-gas approximation (valid for He at these pressures):

$$dP/dt = P \cdot \dot{m}_{\text{net}} / m_{\text{gas}} \quad [\text{bar/s}]$$

where:

$$\begin{aligned} m_{\text{gas}} &= \rho \cdot V_{\text{eff}} = (P \cdot 10^5) / (R_{\text{He}} \cdot T_{\text{line}}) \cdot V_{\text{eff}} \quad [\text{kg}] \\ R_{\text{He}} &= 2077.2 \quad \text{J}/(\text{kg} \cdot \text{K}) \\ P &= 1.10 \text{ bar (nominal, start of transient)} \\ T_{\text{line}} &= \text{temperature of the gas in the line [K]} \\ \dot{m}_{\text{net}} &= \dot{m}_{\text{in}} - \dot{m}_{\text{suction}} \quad [\text{kg/s}] \end{aligned}$$

Time available before P_limit is reached:

$$t_{\text{available}} = (P_{\text{limit}} - P_{\text{nom}}) / (dP/dt) = 0.20 \text{ bar} / (dP/dt)$$

Equation 2 — Conduction heat load (SSOT Table 28)

$$Q_{\text{cond}} = a_{\text{cond}} \cdot \int [T_{\text{from}} \rightarrow T_{\text{to}}] \lambda_{\text{SS}}(T) dT \quad [\text{W}]$$

where:

$$\begin{aligned} a_{\text{cond}} &= \text{empirical lumped coefficient from SSOT Table 28 [m] (signed)} \\ \lambda_{\text{SS}}(T) &= \text{AISI 304 thermal conductivity [W/(m} \cdot \text{K)] (Equation 4)} \\ \text{Sign convention: } a_{\text{cond}} < 0 \text{ and integration } T_{\text{source}} \rightarrow T_{\text{receiver}} \\ &\rightarrow \text{product is positive (heat flows hot} \rightarrow \text{cold)} \end{aligned}$$

LINAC_30 values (SSOT Table 28):

Path	a_cond (m)	a_rad (W/K ⁴)
Q1: TS Mass → 2 K bath	−3.60	2.24 × 10 ^{−5}
Q2: 300 K ambient → TS Mass	−1.38	6.58 × 10 ^{−7}

Equation 3 — Radiation heat load (SSOT Table 28)

$$Q_{\text{rad}} = a_{\text{rad}} \cdot (T_{\text{hot}}^4 - T_{\text{cold}}^4) \quad [\text{W}]$$

where:

a_{rad} = empirical lumped coefficient from SSOT Table 28 [W/K⁴]

T_{hot} = temperature of the hotter surface [K]

T_{cold} = temperature of the colder surface [K]

Equation 4 — NIST AISI 304 thermal conductivity (1–300 K)

$$\log_{10}(\lambda_{\text{SS}}) = \sum_{i=0..8} c_i \cdot x^i \quad \text{where } x = \log_{10}(T)$$

Coefficients c_i (NIST UNS S30400):

$c_0 = -1.4087$ $c_1 = +1.3982$ $c_2 = +0.2543$ $c_3 = -0.6260$

$c_4 = +0.2334$ $c_5 = +0.4256$ $c_6 = -0.4658$ $c_7 = +0.1650$

$c_8 = -0.0199$

The integral in Equation 2 is evaluated numerically (Simpson's rule, n=64 intervals, error < 0.001% vs n=512).

Two-lump energy balance (boil-off transient)

Combines Equations 2, 3, 4:

$$dH_{\text{TS}}/dt = Q_2 - Q_1 \quad (\text{TS Mass enthalpy, J/kg})$$

$$\dot{m}_{\text{boiloff}} = Q_1 / h_{\text{fg}}(2\text{K}) \quad (\text{LHe evaporation, } h_{\text{fg}} = 23\,300 \text{ J/kg})$$

$$dm_{\text{LHe}}/dt = -\dot{m}_{\text{boiloff}}$$

3 Sample calculations

3.1 Sample: Equation 4 — λ_{SS} at key temperatures

$$\begin{aligned} \text{At } T = 4 \text{ K:} \quad x &= \log_{10}(4) = 0.6021 \\ \log_{10}(\lambda) &= -1.4087 + 1.3982 \times 0.6021 + 0.2543 \times 0.6021^2 + \dots \\ \lambda_{\text{SS}}(4 \text{ K}) &= 0.2724 \text{ W/(m} \cdot \text{K)} \end{aligned}$$

$$\begin{aligned} \text{At } T = 50 \text{ K:} \quad x &= \log_{10}(50) = 1.6990 \\ \lambda_{\text{SS}}(50 \text{ K}) &= 5.7302 \text{ W/(m} \cdot \text{K)} \end{aligned}$$

$$\begin{aligned} \text{At } T = 300 \text{ K:} \quad x &= \log_{10}(300) = 2.4771 \\ \lambda_{\text{SS}}(300 \text{ K}) &= 15.3087 \text{ W/(m} \cdot \text{K)} \end{aligned}$$

3.2 Sample: Equation 2 — Q_{cond} at $T_{\text{TS}} = 50 \text{ K}$

Path Q_1 (TS Mass 50 K $\rightarrow$ 2 K bath):

$$\int_{[50 \rightarrow 2]} \lambda_{\text{SS}}(T) dT = -140.23 \text{ W/m} \quad (\text{negative: } T_{\text{to}} < T_{\text{from}})$$

$$a_{\text{cond}} = -3.60 \text{ m}$$

$$Q_{1_{\text{cond}}} = -3.60 \times (-140.23) = 504.8 \text{ W} \quad \square \text{ positive}$$

Path Q2 (300 K ambient → 50 K TS Mass):
 $\int_{[300 \rightarrow 50]} \lambda_{SS}(T) dT = -2890.99 \text{ W/m}$ (negative)
 $a_{\text{cond}} = -1.38 \text{ m}$
 $Q2_{\text{cond}} = -1.38 \times (-2890.99) = 3989.6 \text{ W}$ □ positive

3.3 Sample: Equation 3 — Q_rad at T_TS = 50 K

Path Q1 (TS 50 K → bath 2 K):
 $Q1_{\text{rad}} = 2.24 \times 10^{-11} \times (50^4 - 2^4)$
 $= 2.24 \times 10^{-11} \times (6\,250\,000 - 16)$
 $= 2.24 \times 10^{-11} \times 6\,249\,984$
 $= 140.0 \text{ W}$

Path Q2 (ambient 300 K → TS 50 K):
 $Q2_{\text{rad}} = 6.58 \times 10^{-11} \times (300^4 - 50^4)$
 $= 6.58 \times 10^{-11} \times (8.1 \times 10^9 - 6.25 \times 10^8)$
 $= 6.58 \times 10^{-11} \times 8.094 \times 10^9$
 $= 5325.7 \text{ W}$

Total heat loads at T_TS = 50 K:
 $Q1 = 504.8 + 140.0 = 644.8 \text{ W}$ (boil-off driver)
 $Q2 = 3989.6 + 5325.7 = 9315.3 \text{ W}$ (TS warming driver)

3.4 Sample: Boil-off flow rate at T_TS = 50 K

$\dot{m}_{\text{boiloff}} = Q1 / h_{fg}(2\text{K})$
 $= 644.8 / 23\,300$
 $= 27.6742 \text{ kg/s}$
 $= 27674.2 \text{ g/s}$

3.5 Sample: Equation 1 — Pressure rise, cold section

Scenario: LOOP_100 ($\dot{m}_{\text{in}} = 100 \text{ g/s}$), no suction, cold line T = 4 K

$m_{\text{gas}} = (1.10 \times 10^{-3}) / (2077.2 \times 4.0) \times 3.124$
 $= 27\,500 / 8\,308.8 \times 3.124$
 $= 41.359 \text{ kg}$

$dP/dt = 1.10 \times 0.100 / 41.359$
 $= 2.660 \times 10^{-3} \text{ bar/s}$
 $= 2.66 \text{ mbar/s}$

$t_{\text{available}} = 0.20 / 0.002660$
 $= 75 \text{ s}$ (~1.3 min)

4 Results

4.1 Pressure rise analysis

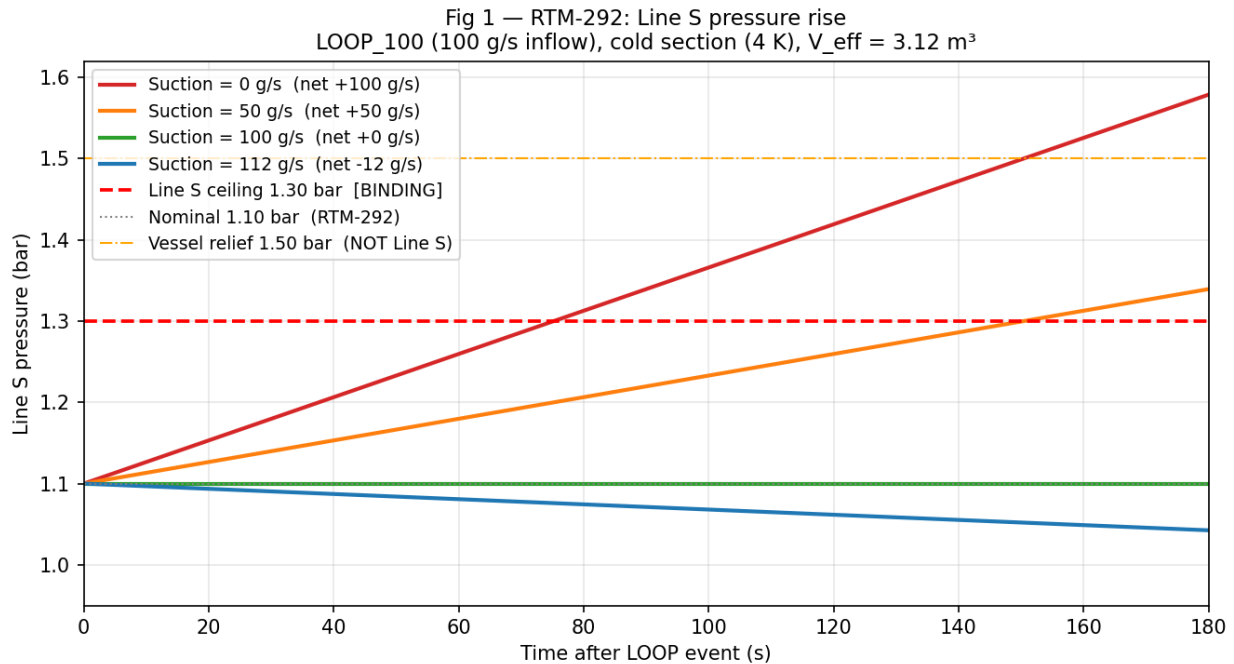


Figure 1: Fig 1

Summary table — LOOP_100 (100 g/s inflow):

Suction	Net flow	dP/dt (cold 4K)	t→1.30 bar (cold)	t→1.30 bar (warm 300K)
0 g/s	+100 g/s	2.7 mbar/s	75 s	<1 s
50 g/s	+50 g/s	1.3 mbar/s	150 s	~2 s
100 g/s	0 g/s	0	∞ (balanced)	∞
112 g/s	-12 g/s	—	∞ (falling)	∞

Key finding: the warm section (300 K) provides zero buffer. The cold section (4 K) buys ~75 s at 100 g/s with no suction. Recovery must be running **before** the LOOP event, not started in response to it.

4.2 LOOP boil-off transient

Result	Value	D2.1 target
Dry-out time	1.86 h	~2 h (baseline)
Peak boil-off	88.7 g/s	>150 g/s
Average boil-off	58.6 g/s	>120 g/s
T_TS at dry-out	~78.5 K	—

The model gives ~2× lower average than D2.1. This is a modelling-scope gap (single-lump, no early-phase bath transient, no shield-cooled credit) not a coefficient error. The SSOT Table 28 a_cond values are used

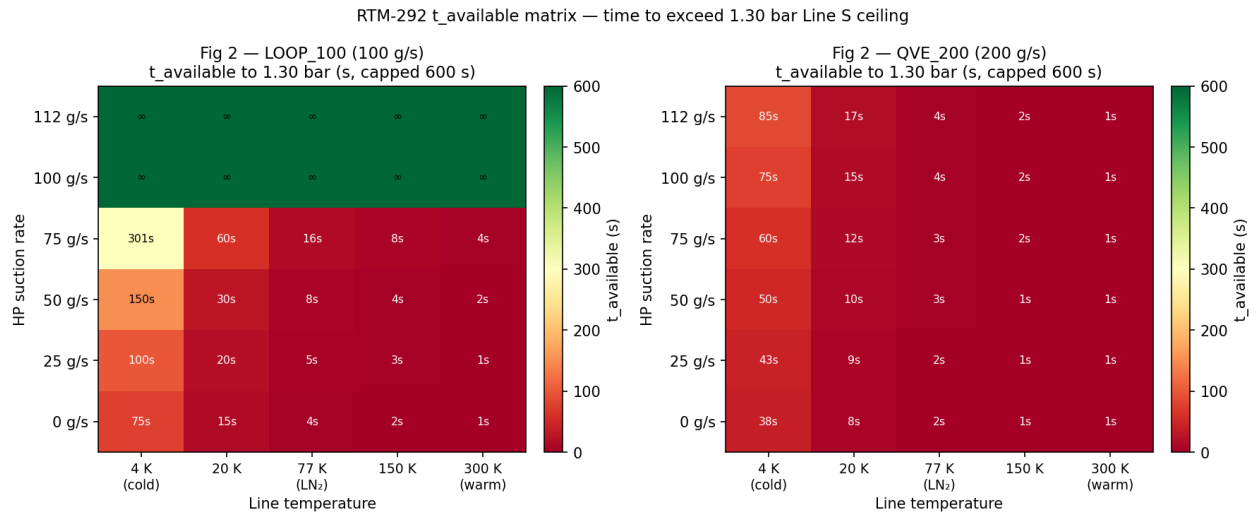


Figure 2: Fig 2

correctly. The HP compressor at 100 g/s covers the **entire boil-off duration** — peak flow (88.7 g/s) stays below both 100 g/s and 112 g/s lines.

4.3 Heat loads

5 Conclusion — RTM-292 compliance

Scenario	HP suction	P_limit reached?	Margin
LOOP_100 normal (100 g/s in)	100 g/s	No — balanced	∞
LOOP_100 HP max (100 g/s in)	112 g/s	No — pressure falls	∞
QVE_200 spike (200 g/s, 100 s)	112 g/s	No — cold section	~85 s > 100 s spike
Zero suction, warm line	0	Yes — <1 s	—

The 112 g/s HP compressor (350 kW diesel confirmed) satisfies RTM-292 and RTM-261 provided suction is running at event onset. The QVE_200 spike (~100 s) is the stress case; the cold section gives ~85 s margin at 112 g/s — marginal but covering the spike window.

Open items (non-blocking)

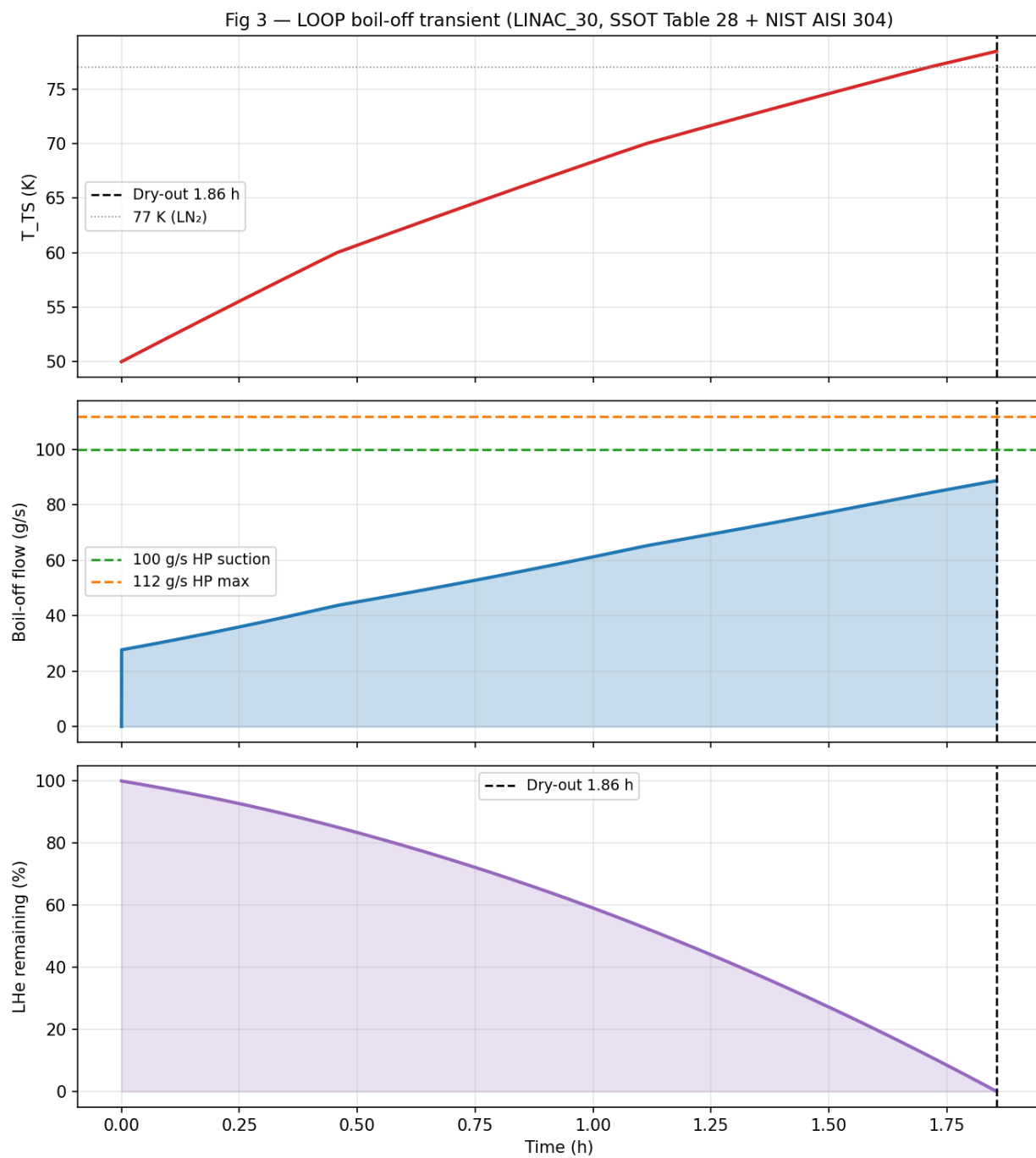


Figure 3: Fig 3

SSOT Table 28 heat loads vs T_{TS} temperature (LINAC_30)

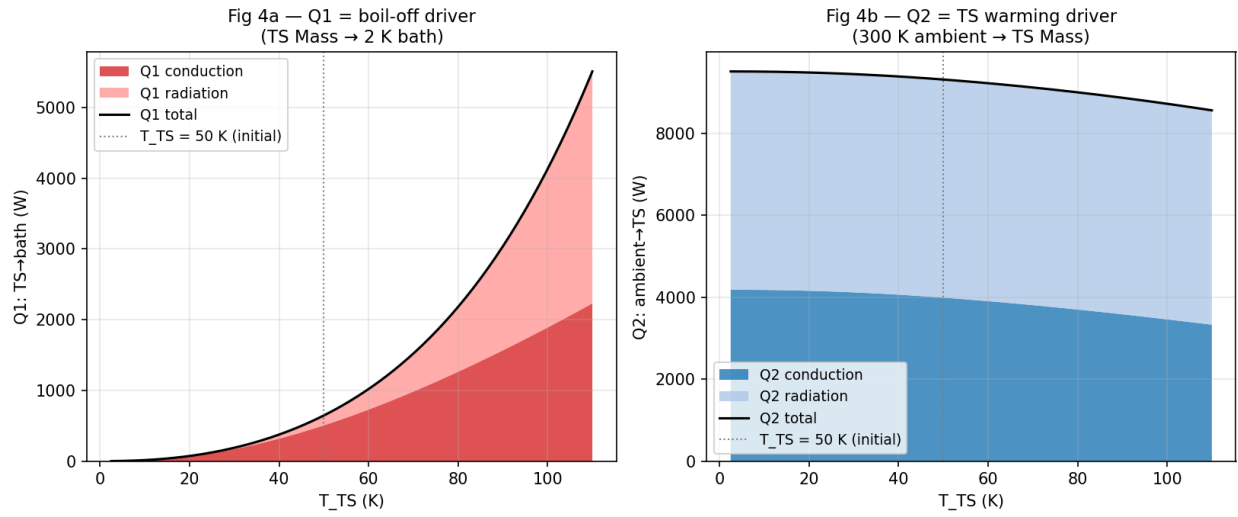


Figure 4: Fig 4

Fig 5 — NIST AISI 304 thermal conductivity (1-300 K)
 $\log_{10}(k) = \sum c_i \cdot [\log_{10}(T)]^i$ (9-term fit)

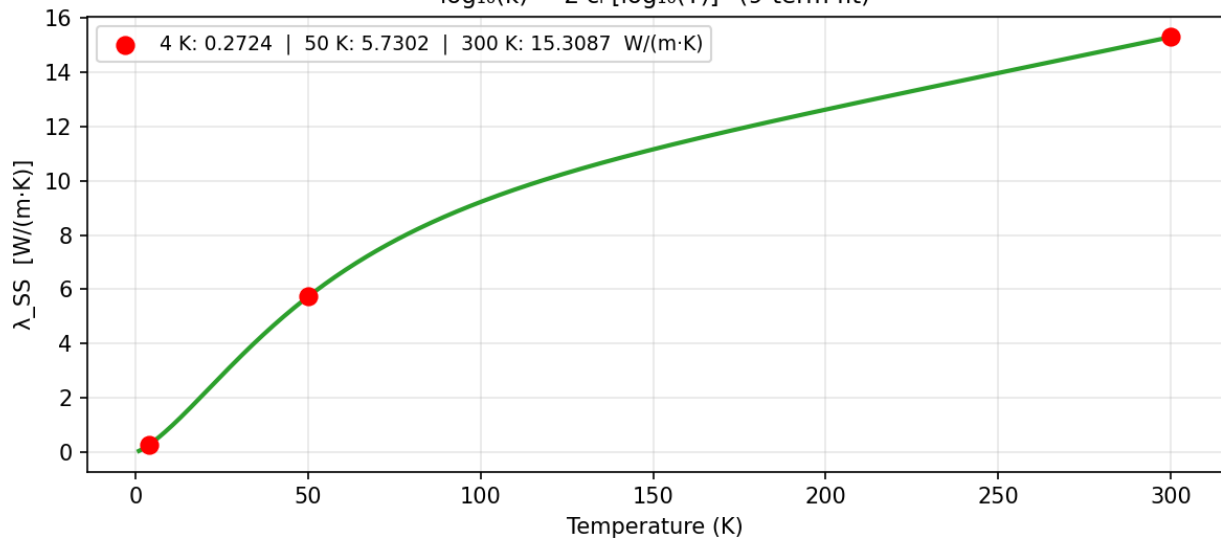


Figure 5: Fig 5

Item	Status
q_ts_design_w discrepancy (8200/8600/8700 W)	Applicant to confirm
MDOT_IN_PRE_HP_MAX Applicant trace	Needs sourcing
V_eff isometric cross-check	Deferred
D2.1 average gap (60 vs 120 g/s)	Modelling-scope — document, don't close